BuildBridge SA

Project Report

INF3014F 2026

Bridging the gap between construction surplus and community development

Course: INF3014F
Due Date: 20 April 2026

1. Introduction

This idea is rooted from the "Sustainability & Circular Economy" and "Township and Informal Economy" focus areas required by the brief.

In the South African construction sector, a critical market inefficiency exists: large-scale civil engineering projects routinely generate vast quantities of usable surplus materials, which range from structural bricks and cement to timber and roofing, which are financially burdensome to store, transport, or dispose of. Simultaneously, within the township and informal economies, community builders and DIY home improvers are severely bottlenecked by the inflated retail costs of these exact materials.

BuildBridge SA is a localized B2B-to-C e-commerce marketplace designed to bridge this exact gap. By functioning as a digital clearinghouse, the platform allows civil contractors in the Western Cape to list surplus materials at 50–80% below standard retail prices. This model directly addresses a real-world social challenge by transforming expensive construction waste into an accessible community opportunity, establishing a viable, socially impactful e-business model.

Developed from scratch as a responsive, high-performance React Application, BuildBridge SA transcends standard e-commerce functionality. To cater to informal builders who may lack formal quantity surveying expertise, the platform features a custom-built "Smart Project Estimator." This innovative tool programmatically calculates material requirements based on user-inputted physical dimensions and dynamically curates "Surplus Bundles" from the real-time database. By marrying scalable web infrastructure with intuitive user experience design, BuildBridge SA provides a frictionless, transactional solution that drives measurable social impact while fostering a localized circular economy.

2. Business Model Canvas

BuildBridge SA Business Model Canvas

BuildBridge SA is a localized B2B-to-C surplus construction materials marketplace.

Key Partners

- Large civil engineering and construction firms (supply side).
- Local township logistics providers ("Bakkie" owners).
- NGOs focused on informal settlement upgrading.
- Western Cape Government / DEADP.

Key Activities

- Platform development and maintenance
- Supplier vetting and material quality verification
- Facilitating local logistics.
- Community outreach and supplier acquisition.

Key Resources

- The eCommerce platform and Smart Estimator algorithm
- Surplus materials database
- Supplier and buyer network
- Brand and community trust

Value Proposition

Suppliers:

- Zero-cost or low-cost disposal of usable surplus materials
- ESG compliance contribution
- Rapid clearance

Buyers:

- Access to quality structural materials for cheaper
- The Smart Project Estimator
- Transparent pricing/no hidden fees:

Customer Relationships

- Automated self-service via the web platform
- Community trust-building through transparent pricing
- WhatsApp community integration

Channels

- Direct B2B outreach to construction site managers.
- Social media
- referrals
- BuildBridge SA website

Customer Segments

Suppliers (B2B):

- Medium-to-large civil engineering and construction companies

Buyers (B2C):

- DIY home improvers in townships.
- Informal builders/contractors.
- Spaza shop owners.

Cost Structure

- Web hosting and platform infrastructure.
- Platform development: Initial development cost; ongoing maintenance and feature development.
- Marketing and community outreach.
- Logistics coordination overhead: Managing "bakkie" delivery partnerships.
- Customer support.

Revenue Streams

- 7.5% transaction fee charged to the buyer at checkout.
- "Premium Clearance" listing fee charged to contractors.
- Monthly subscription tiers for high-volume suppliers who regularly list materials.

2.1 Key Partners

- Large civil engineering and construction firms (supply side): Companies completing commercial builds in the Western Cape generate significant surplus materials that must be cleared from site within contractual deadlines.
- Local township logistics providers ("Bakkie" owners): Last-mile delivery is provided by informal logistics operators using small trucks, a well-established informal economy sector in Cape Town.
- NGOs focused on informal settlement upgrading: Organisations such as Habitat for Humanity South Africa provide community trust and potential bulk procurement partnerships.
- Western Cape Government / DEADP: Environmental compliance alignment with the Department of Environment and Development Planning strengthens the ESG value proposition for contractor partners.

2.2 Key Activities

- Platform development and maintenance: Ongoing development of the React-based web application including the Smart Project Estimator, product catalog, cart, and checkout flow.
- Supplier vetting and material quality verification: Digital verification process for contractor-listed materials including condition grading and photographic documentation.
- Facilitating local logistics: Coordinating "bakkie" delivery partners with buyer delivery addresses using location data.
- Community outreach and supplier acquisition: Direct B2B outreach to construction site managers and digital community engagement through WhatsApp groups and Facebook township community pages.

2.3 Key Resources

- The eCommerce platform and Smart Estimator algorithm: The core technical asset — a custom JavaScript tool that calculates exact material quantities from physical dimensions and auto-matches available surplus stock.
- Surplus materials database: A curated, real-time inventory of available materials from verified contractor partners across the Western Cape.
- Supplier and buyer network: Relationships with construction firms (supply) and community builders (demand) form the core network effect asset of the business.
- Brand and community trust: Transparent pricing, honest condition grading, and social impact positioning build the trust required for informal sector buyers to transact digitally.

2.4 Value Propositions

For Contractors (Suppliers):

- Zero-cost or low-cost disposal of usable surplus materials that would otherwise incur dump fees of R800–R2,500 per load (City of Cape Town refuse disposal tariffs, 2024/25).
- ESG compliance contribution: Documented diversion of waste from landfill supports contractors' Environmental, Social, and Governance reporting — increasingly required for government tender eligibility.
- Rapid clearance: "Premium Clearance" listing option guarantees materials are moved off-site within 24 hours.

For Community Builders (Buyers):

- Access to quality structural materials at 50–80% below standard retail prices (benchmarked against Builders Warehouse and Cashbuild, Western Cape, 2024).
- The Smart Project Estimator: A free digital quantity surveying tool that calculates exactly how many bricks, cement bags, and cubic metres of sand are needed for a given wall — eliminating costly over-purchasing errors.
- Transparent pricing: A single 7.5% platform fee disclosed at checkout — no hidden charges.

2.5 Customer Relationships

- Automated self-service via the web platform: Buyers can browse, estimate, add to cart, and checkout without human intervention.
- Community trust-building through transparent pricing: All prices, condition grades, and stock quantities are displayed honestly.
- WhatsApp community integration: Activity notifications and material alerts distributed through existing WhatsApp township community groups.

2.6 Channels

- Direct B2B outreach to construction site managers: Personal selling to project managers nearing project completion, when surplus clearance becomes urgent.
- Social media — Facebook Groups and WhatsApp community groups: Township Facebook groups are primary discovery channels for buyer-side users.
- Word of mouth within builder networks: Informal builders in townships operate within tight community networks — positive experiences drive organic referrals.
- The BuildBridge SA website: Direct channel for browsing, estimating, and transacting.

2.7 Customer Segments

Suppliers (B2B):

Medium-to-large civil engineering and construction companies operating in the Western Cape. Specifically targeting project managers and site managers of companies completing commercial or residential builds with contract-end deadlines for site clearance.

Buyers (B2C):

- DIY home improvers in townships: Residents of Khayelitsha, Mitchells Plain, Gugulethu extending their homes or building boundary walls.
- Informal builders/contractors: Individuals hired by township residents to undertake small building projects — the primary user persona.
- Spaza shop owners: Small business owners seeking affordable materials for structural improvements.

2.8 Cost Structure

- Web hosting and platform infrastructure: Cloud hosting costs (estimated R200–R500/month at scale).
- Platform development: Initial development cost; ongoing maintenance and feature development.
- Marketing and community outreach: Social media advertising targeted at Cape Town townships; direct sales effort for contractor acquisition.
- Logistics coordination overhead: Managing "bakkie" delivery partnerships.
- Customer support: WhatsApp-based support for buyers unfamiliar with digital commerce.

2.9 Revenue Streams

- 7.5% transaction fee charged to the buyer at checkout: Applied to every completed transaction. At an average order value of R1,500, this generates R112.50 per transaction.
- "Premium Clearance" listing fee charged to contractors: A fixed fee (R350–R800 per listing) for contractors requiring guaranteed 24-hour site clearance.
- Future — Volume subscription for large contractors: Monthly subscription tiers for high-volume suppliers who regularly list materials across multiple project sites.

How I Derived Facts and Assumptions

Construction Waste Disposal Costs (R800–R2,500 per load):

Based on the City of Cape Town's published refuse and waste disposal tariffs for 2024/25, which charge commercial construction waste by volume. Private skip hire companies in Cape Town quote R900–R2,200 for a standard 6m^3 skip. Source: City of Cape Town Municipal Tariffs 2024/25.

Retail Price Benchmark (50–80% below retail):

Benchmarked against Builders Warehouse (Parow) and Cashbuild (Bellville) online price lists: clay facebrick at R2.80–R3.50/brick; OPC cement at R89–R110/bag; structural timber at R45–R65/m. Source: Builders Warehouse ZA and Cashbuild ZA websites, April 2025.

Platform Fee (7.5%):

Modelled on comparable South African marketplace platforms: Takealot charges 8–15% category commissions; smaller niche platforms charge 5–10%. At 7.5%, BuildBridge SA is positioned at the lower end to encourage adoption in a price-sensitive market. Source: Takealot Marketplace Seller Agreement, 2024.

3. User Experience Analysis

3.1 User Personas

BuildBridge SA serves two distinct user groups: community builders who need affordable materials, and civil contractors who need to clear surplus stock. The following personas were derived from the target market and directly informed every major design decision on the platform.

Persona 1: Sedzesani Mudau (The Buyer):

USER PERSONAS

SEDZESANI MUDAU

Age: 35
Occupation: Informal builder/subcontactor
Location: Khayelitsha, Cape Town
Education: Grade 12

BACKGROUND

Sedzesani has been building walls, extending shack rooms into brick structures, and repairing roofing across Khayelitsha for eight years. He gets most of his work through word of mouth and WhatsApp referrals. He is typically hired by homeowners who want to expand their informal dwellings or repair storm damage. Sipho does not own a large vehicle, so he relies on local bakkie drivers for material delivery.

GOALS

- Source affordable, quality construction materials without travelling.
- Calculate exactly how many bricks and bags of cement he needs so he does not over-buy.
- Get materials delivered to the job site, not to a depot
- Build a reputation for completing jobs on budget and on time.

PAIN POINTS

- Retail hardware stores (Builders Warehouse, Build It) charge full price, the margins eat into his earnings.
- He often miscalculates material quantities and has to make multiple trips, costing him data and taxi fare
- Untrustworthy suppliers with unknown stock levels waste his time.
- He is not always comfortable with technical language or complex interfaces on websites.

QUOTE

"I need to know exactly what to buy before I go anywhere. Getting it wrong costs me jobs."

TECHNOLOGY COMFORT

- Comfortable with WhatsApp, Facebook, and basic mobile browsing.
- Not comfortable with multi-step account registration or complex navigation.
- Prefers visual content (I mages, icons) over dense text.

HOW BUILDBRIDGE SA SERVES SEDZESANI

- The Smart Project Estimator lets him input wall dimensions and get exact material quantities, removing the guesswork that causes overbuying.
- Prices are shown upfront in Rands per unit with clear location info
- The 10% bundle discount rewards buying the full recommended bundle in one click
- Cart and checkout are simple and visual with minimal form fields

Persona 2: Gift Maluleke (The Supplier):

USER PERSONAS

GIFT MALULEKE

Age: 42
Occupation: SIte Manager
Location: Milnerton, Cape Town
Education: Degree in Construction management

GOALS

- Clear surplus materials quickly
- Avoid paying landfill or skip fees for usable material
- Record the disposal in a way that contributes to the firm's ESG reporting
- Not spend more than 20 minutes setting up a listing

QUOTE

"I need the excess materials gone by Friday. Every day they sit here is money wasted."

BACKGROUND

Gift manages a team of 40 on a commercial office park development. They are two weeks from handover. The site has leftover stock which are three pallets of facebrick, 50 bags of OPC cement, and a bundle of treated timber offcuts. Disposing of this material legally costs money and requires booking a skip. His company is also under pressure to improve its ESG score before its next annual report.

PAIN POINTS

- Traditional disposal methods (skips, landfill) cost money and generate paperwork
- He cannot justify the admin overhead of listing items on Facebook Marketplace individually
- His company's compliance team requires proof of responsible disposal
- He will only trust a platform that looks legitimate and professional

TECHNOLOGY COMFORT

- Comfortable with WhatsApp, email, and project management apps.
- Confident with multi-step registrations and online portals.
- Skilled in mobile and desktop browsing.
- Uses construction apps like BIM viewers and AutoCAD mobile.
- Relies on digital tools for scheduling, cost tracking, and resource allocation.
- Not comfortable with coding or advanced IT troubleshooting.

HOW BUILDBRIDGE SA SERVES GIFT

- Split hero on the landing page immediately identifies the platform serves both builders and contractors.
- Zero listing fee removes the financial barrier to participation.
- ESG credit mechanism gives Gift a digital record of responsible surplus disposal.
- Platform fee is charged to the buyer, Gift pays nothing to list.

3.2 Hierarchical Task Analysis (HTA)

3.2.1 Smart project estimator

Task:
Sedzesani uses the Smart Project Estimator to purchase materials for a 5-metre boundary wall.

Goal:
Complete the purchase of the correct quantity of bricks and cement at the lowest available price, with delivery to his job site.

0. Purchase materials for a 5-metre boundary wall

using the BuildBridge SA Smart Estimator

Plan 0. Do 1 → 2 → 3 → 4 in sequence. If stock is unavailable at step 3.2, go to step 3.3 (browse shop manually). If checkout fails, retry step 4.

1. Access the Smart Project Estimator

- 1.1 Open BuildBridge SA on mobile browser
- 1.2 Observe landing page hero section
- 1.3 Tap "Smart Estimator" in nav
  - Alt: tap CTA button on landing page
- 1.4 Estimator page loads with two sliders (Wall Length, Wall Height)

2. Input project parameters

- 2.1 Drag Wall Length slider to 5 m
  - 2.1.1 Live wall diagram updates proportionally
  - 2.1.2 Value below slider reads "5m"
- 2.2 Drag Wall Height slider to 1.8 m
  - 2.2.1 Live wall diagram updates
  - 2.2.2 Value reads "1.8m"

3. Review output & select bundle

- 3.1 Read result cards: Wall Area 5 m^2, Bricks 450, Cement 3 bags, 9 m^3
- 3.2 View Recommended Bundle panel
  - 3.2.1 Read bricks line item name & qty
  - 3.2.2 Read cement line item name & qty
  - 3.2.3 Read "After 10% bundle discount" price
- 3.3 (If no stock) Read error → navigate to /shop manually
- 3.4 Tap "Add Bundle to Cart (10% Off)"
  - 3.4.1 Button turns green "✓ Bundle Added to Cart!" 2s
  - 3.4.2 Cart icon updates with item count

4. Complete checkout

- 4.1 Tap cart icon → open Cart Drawer
  - 4.1.1 Verify line items match bundle
  - 4.1.2 Verify 10% bundle discount deduction
  - 4.1.3 Verify 7.5% platform fee
  - 4.1.4 Verify final total in Rands
- 4.2 Tap "Proceed to Checkout →"
- 4.3 Complete checkout form (name, address, contact)
- 4.4 Tap "Complete Order"
- 4.5 Observe success screen with confetti
- 4.6 Note order reference for WhatsApp follow-up

HTA Summary Table:

Step	Action	Trigger	Feedback
1.3	Tap nav link	User intent	Page transition, scroll to top
2.1	Drag length slider	Touch gesture	Diagram updates live
2.2	Drag height slider	Touch gesture	Diagram updates live
3.4	Tap bundle button	User decision	Button turns green, cart updates
4.2	Tap checkout	User intent	Checkout page loads
4.4	Tap complete order	User confirmation	Confetti, success message

3.2.2 General Website Use (Browse, Add to Cart, Checkout)

Task:
A community builder visits BuildBridge SA to browse available materials, add items to cart, and complete a purchase without using the Smart Estimator.

Goal:
Find a suitable product, review its details, add it to the cart, and successfully place an order.

0. Purchase surplus construction materials

via BuildBridge SA marketplace

Plan 0. Do 1 → 2 → 3 → 4 → 5 in sequence. If the desired product is not found at step 2, try 2.3 (change filter). If the cart is empty at step 4, return to step 2.

1. Land & orient

- 1.1 Open site in browser
- 1.2 View hero section (Builders / Contractors)
- 1.3 Read impact stats
- 1.4 Browse category tiles
- 1.5 View featured listings

2. Search & filter

- 2.1 Click "Shop" in nav
- 2.2 Marketplace page loads
- 2.3 Tick category filter
  - 2.3.1 Grid updates
  - 2.3.2 Result count shows
- 2.4 Type search keyword
  - 2.4.1 Live filter on keystroke
  - 2.4.2 Non-matches hide
- 2.5 Clear search field

3. Review & add to cart

- 3.1 Identify product card
- 3.2 Read name, location, price, badge
- 3.3 (Opt) click heart → wishlist
  - 3.3.1 Heart fills orange
  - 3.3.2 Wishlist count updates
- 3.4 Click "Add to cart"
  - 3.4.1 "✓ Added" shown 1.5s
  - 3.4.2 Cart icon increments
- 3.5 Repeat for more products

4. Review cart & discount

- 4.1 Open cart drawer
- 4.2 Review items list
  - 4.2.1 Qty, price, remove shown
- 4.3 Adjust quantity (±)
  - 4.3.1 Totals update live
- 4.4 Review pricing breakdown
  - 4.4.1 Subtotal
  - 4.4.2 Bundle −10%
  - 4.4.3 Platform fee +7.5%
  - 4.4.4 Total in rands
- 4.5 Click "Proceed to checkout →"

5. Complete checkout

- 5.1 Checkout page loads
- 5.2 Fill delivery details
  - 5.2.1 Full name
  - 5.2.2 Delivery address
  - 5.2.3 Contact number
- 5.3 Review order summary
- 5.4 Click "Complete order"
- 5.5 Confetti + success message
- 5.6 Order confirmed

HTA 2 Summary Table:

Step	Action	Trigger	Feedback
1.4	Scroll to categories	User explores	Category tiles visible
2.1	Click Shop nav link	User intent	Shop page loads, 12 products shown
2.3	Click filter checkbox	User selects category	Grid updates instantly, count changes
2.4	Type in search bar	User input	Grid filters live on each keystroke
3.3	Click heart icon	User saves item	Heart fills orange, wishlist count updates
3.4	Click Add to Cart	User decision	Button turns green, cart count increments
4.3	Adjust quantity	User updates order	Line total and grand total update instantly
5.4	Click Complete Order	User confirms	Confetti animation, success message shown

4. Creativity & Ingenuity: The Smart Project Estimator

Feature Name: The Smart Project Estimator & Surplus Bundler

Standard e-commerce requires users to know exactly what they want. In the informal building sector, lack of technical knowledge leads to purchasing errors — buying too little cement, or too many bricks. This feature acts as a digital Quantity Surveyor.

The estimator uses custom JavaScript logic to take physical dimensions (wall length and height), calculate standardized material needs using standard South African bricklaying ratios, and programmatically query the product database to build a custom "Surplus Bundle" that the user can purchase with a single click at a 10% discount.

The calculation formula:

- Wall Area = Length × Height (in m^2)
- Bricks Required = ceil(Area × 50) — standard 50 bricks per m^2 ratio
- Cement Bags = ceil(Bricks / 150) — 1 bag per 150 bricks for standard mortar mix
- Sand = ceil(Area × 0.02 × 100) / 100 — 0.02 m^3 of sand per m^2 of wall

Why this cannot be replicated on a standard CMS:

Platforms like Wix or Shopify cannot dynamically calculate physical volume formulas and auto-generate cart bundles based on variable user inputs against live inventory. This requires custom JavaScript code and a dynamic product-matching algorithm — features that are uniquely coded for this platform and not available as off-the-shelf templates or plugins.

5. Wireframes

All wireframes are low-fidelity, showing layout and structure using grey placeholder boxes. They show the core e-commerce features: search, product grid, cart, wishlist, pricing, discounts, and checkout flow.

WIREFRAME — BUILDBRIDGE SA — LANDING PAGE (/)

NAVBAR

LOGO: BuildBridge SA

Home | Shop | Smart Estimator | About | Contact

☾ ♡ 🛒 Cart

HERO SECTION — SPLIT LAYOUT

For Builders

Browse Materials →

For Contractors

List Materials →

IMPACT STATS

2,400+	R1.2M	340+
Tonnes Diverted from Landfill	Saved by Community Builders	Community Builders Helped

BROWSE BY CATEGORY

Bricks | Cement | Timber | Roofing | Scaffolding

FEATURED LISTINGS

[Product Image] — Add to Cart

[Product Image] — Add to Cart

[Product Image] — Add to Cart

HOW IT WORKS

1. Browse & Estimate
2. Add to Cart
3. Checkout & Deliver

FOOTER

BuildBridge SA | Navigate | Impact

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WIREFRAME — BUILDBRIDGE SA — SHOP / PRODUCT CATALOG (/SHOP)

NAVBAR

LOGO: BuildBridge SA

Home | Shop (active) | Smart Estimator | About | Contact

Surplus Marketplace

12 listings available across Cape Town

SEARCH BAR

Search materials...

FILTER SIDEBAR

CATEGORY

- [x] Bricks
- [ ] Cement
- [ ] Timber
- [ ] Roofing
- [ ] Scaffolding

Clear All Filters

PRODUCT GRID

[Product Image] Add

[Product Image] Add

[Product Image] Add

CART DRAWER (SLIDES IN FROM RIGHT)

Page content (dimmed)

Shopping Cart

Product × qty | Price | Remove

Product × qty | Price | Remove

Total R [total]

Proceed to Checkout →

FOOTER

FOOTER — Same as Landing Page

WIREFRAME — BUILDBRIDGE SA — SMART PROJECT ESTIMATOR (/ESTIMATOR)

NAVBAR

LOGO: BuildBridge SA

Home | Shop | Smart Estimator (active) | About | Contact

Smart Project Estimator

Calculate exactly what you need before you buy.

INPUT FORM

Wall Length

1m — 20m

Wall Height

0.5m — 5m

LIVE WALL DIAGRAM

Live wall preview (SVG)

CALCULATION RESULTS

BRICKS REQUIRED	CEMENT BAGS
— bricks	— bags (50kg)
SAND REQUIRED	**TOTAL AREA**
— cubic metres	— m² wall surface

Recommended Bundle

[Bricks product] × qty

[Cement product] × qty

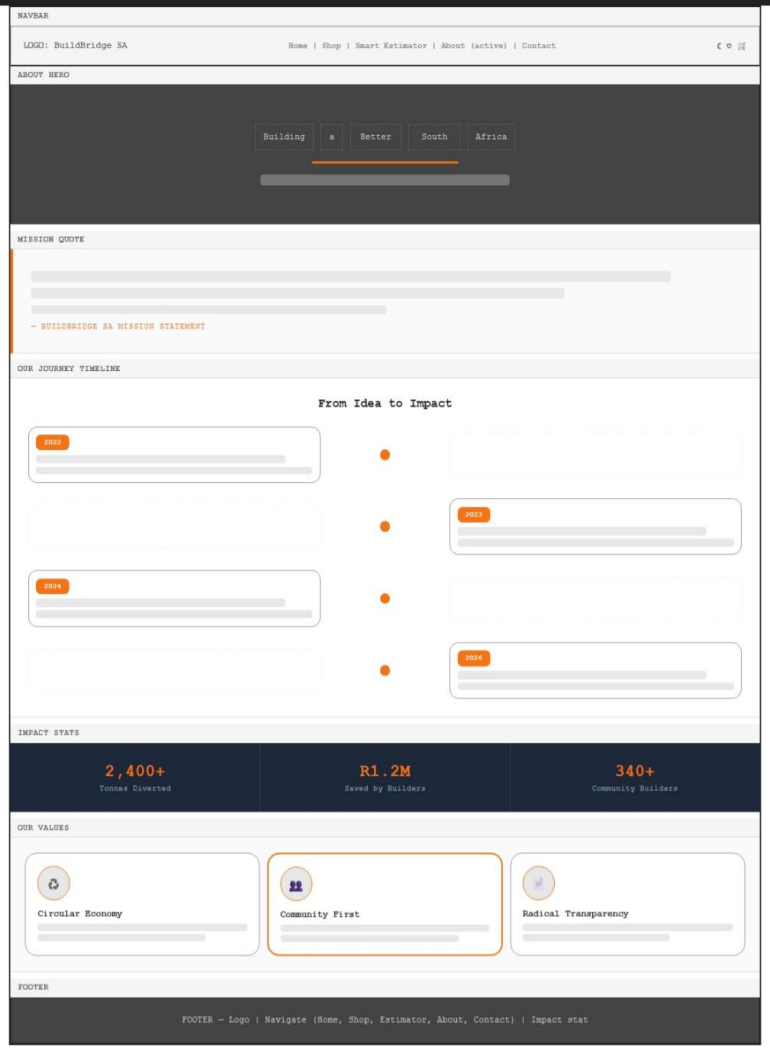
After 10% bundle discount: R [total]

Add Bundle to Cart (10% Off)

FOOTER

FOOTER — Same as Landing Page

WIREFRAME — BUILDBRIDGE SA — ABOUT PAGE (/ABOUT)



WIREFRAME — BUILDBRIDGE SA — CONTACT PAGE (/CONTACT)

NAVBAR

LOGO: BuildBridge SA

Home | Shop | Smart Estimator | About | Contact (active)

CONTACT HERO

GET IN TOUCH

Let's Build Together

CONTACT LAYOUT

REACH US

EMAIL

WHATSAPP

LOCATION

OFFICE HOURS

SEND A MESSAGE

Full Name

[Input field]

Email Address

[Input field]

Subject

[Select field]

Your Message

[Textarea]

Send Message

FOOTER

FOOTER — Logo | Navigate (Home, Shop, Estimator, About, Contact) | Impact stat

6. AI Usage Declaration

AI tools (specifically Claude Code by Anthropic) were used as a development assistant in this project. I directed the AI to implement specific components and features based on the student's own design decisions, business logic, and creative direction. The AI acted as a coding helper, it did not independently determine what to build, how to structure the business, or what the UX should look like.

All business ideas, persona details, value propositions, the choice of the Smart Estimator as the creativity feature, and the overall design direction originated with the student. The AI was used to accelerate implementation of React components, Tailwind CSS styling, and test writing, in the same way any student might use a more capable code autocomplete tool.

A full AI Declaration Form is included in the Docs/ folder of the submitted project.